



Openote

PRODUCT & DESIGN OVERVIEW · PLANNING PHASE

OneNote's experience. Obsidian's openness. Every platform.

An open-source, natively cross-platform notebook — freeform canvas, handwriting and math — that never locks your notes inside a proprietary format or a single vendor's cloud.

Your notes. Your format. Every platform.

Draft v0.2 · 22 July 2026

Prepared for the Openote project
Windows · macOS · Linux · tablets

The best freeform notebook is also the most locked-in

Microsoft OneNote is, for a large class of users, the best freeform note-taking tool ever made. Its infinite canvas, notebook hierarchy, handwriting, and math entry remain genuinely unmatched. And yet it carries structural problems Microsoft has shown no interest in fixing — because they are not bugs, they are the business model.

Your notes are trapped

The `.one` / `.onepkg` format is a binary revision store that, in practice, only OneNote can read. On the free tier you can't even keep a purely local copy — your data lives on OneDrive by default. Getting notes *out*, cleanly and with ink intact, ranges from painful to impossible.

Tethered to one ecosystem

No native Linux client at all. Feature parity is uneven even across supported platforms — page templates, find-and-replace, password protection and local notebooks each go missing somewhere.

Core features paywalled

Solving and step-by-step math require an Enterprise or Education subscription. Local notebooks need a paid Office license. AI features need a Copilot license.

Requests never arrive

Native Markdown, an open format, backlinks, syntax-highlighted code, a Linux client — perennial, well-documented requests Microsoft has repeatedly declined.

The gap in one sentence. There are excellent open note apps, excellent open canvases, and excellent open ink apps — but **no open, cross-platform app combines all four of: a freeform infinite canvas, first-class pen and math, a real notebook hierarchy, and an open, durable file format.** That intersection is exactly where OneNote lives, and exactly where it's most vulnerable.

A notebook you never want to leave — and always could

"Your handwritten notes, diagrams and equations are yours — readable a decade from now, editable on any device and any OS, in a format anyone is free to build on. Openote earns its place not by locking you in, but by being good enough that you never want to leave."

Six principles that decide every trade-off

- 01 Your data is yours.** The file format is open, documented and versioned from day one. Openness is a constraint we design under, not a feature we add later.
- 02 Local-first, cloud-optional.** Fully functional offline, on one device, with no account. Sync is something you turn *on*, not something you must escape.
- 03 The canvas is sacred.** Click anywhere, place anything, drag it where it belongs. Panning, zooming and low-latency ink are first-class performance requirements.
- 04 Interpret, don't interrupt.** Markdown renders in place as you write it; math builds up from linear input as you type. No mode-switching for everyday writing.
- 05 Native feel on every platform.** Cross-platform, but not lowest-common-denominator. The pen must feel like a pen; shortcuts must feel native.
- 06 Open by construction.** An open-source license, a publicly specified format, and an architecture others can extend.

POSITIONING

Where Openote stands

Our two nearest rivals each concede half the field. OneNote owns the experience but not the openness. The FOSS Markdown crowd owns the openness but not the canvas-and-ink experience. AFFiNE comes closest architecturally but has no real handwriting and an open-*core* (not fully open) server. **Nobody is standing where we intend to stand.**

Capability	Openote	OneNote	Obsidian	AFFiNE	Joplin
Freeform infinite canvas	✓	✓	plugin	✓	✗
Notebook / section / page hierarchy	✓	✓	folders	workspace	✓
First-class pen / handwriting	✓	✓	✗	✗	✗
Math entry & rendering	✓	paywalled solve	LaTeX	LaTeX	LaTeX
Inline-rendered Markdown	✓	✗	✓	blocks	✓
Open, documented format	✓	impractical	✓	open-core	✓
Local-first, cloud-optional	✓	✗	✓	✓	✓
Native Linux client	✓	✗	✓	✓	✓
Fully open-source app	✓	✗	freeware	open-core	✓

✓ first-class · **amber** partial/caveated · ✗ none. Our position, in one line: **OneNote's experience, Obsidian's openness, on every platform.**

Six product pillars



Freeform Canvas

An unbounded, pannable/zoomable page. Click anywhere to drop a text box; drag it anywhere. Free-form *or* snap-to-grid placement to keep pages tidy.



Structured Notebooks

The hierarchy you know — notebooks, section groups, sections, pages, subpages — with a navigator that scales to thousands of pages.



Expressive Content

Rich text with inline-rendered Markdown, beautiful math entry, responsive pen & ink, **live page embeds** (render part of another page inside this one, always current), images, tables, and syntax-highlighted code.



Open & Local

An open, documented, versioned file format. Local-first, no size limits, crash-safe, inspectable. Clean import/export — including a OneNote importer.



Sync & Collaboration

Optional, conflict-free cross-device sync (bring your own backend), with real-time collaboration on the same foundation later.



Platform & Craft

Desktop-first — Windows, macOS, **Linux** — with tablets close behind. Performance, accessibility and polish as requirements.

The math & ink that set OneNote apart — matched with open tech



Type it, watch it build

A OneNote-style linear syntax where `\sum`, `/`, `^`, `_`, `\int` autoformat into summations with limits, fractions, integrals and matrices — interpreted as you type. Stored as canonical LaTeX, exportable to MathML, always re-editable. *Solving stays a non-goal — we make equations beautiful to write, not to solve.*



Ink that feels like ink

Pressure-sensitive, responsive strokes you can write and draw anywhere, with palm rejection. Stored losslessly as open stroke data (InkML-interchangeable), independent of any recognition. Handwriting-to-text and ink-to-math come later — and per our priorities, we never trade startup speed or consistency for the last few milliseconds of pen latency.

An MVP that is unmistakably Openote

The MVP is the smallest build that is genuinely useful to a switcher on day one: **desktop-first, single-device, local-only, no sync**. The success test says it best —

MVP success test. A OneNote user installs Openote on **Linux**, recreates a typical page — mixed typed text, a hand-drawn diagram, and a summation equation, freely arranged — saves it as an open file on their own disk, and reopens it cleanly. *Nothing about that flow required a Microsoft account, a subscription, or a supported-OS list that excludes them.*

In the MVP

- Freeform canvas: pan/zoom both axes, click-to-create text, drag/resize, free + snap-to-grid
- Notebook / section / page / subpage hierarchy + navigator
- Rich text with inline-rendered Markdown & reveal-on-edit
- Math blocks: linear-builds-to-2D, LaTeX storage, complex notation
- Ink: pen / highlighter / eraser, pressure, palm rejection, lossless
- Images, file attach, syntax-highlighted code
- The open format v1 + Markdown/PDF export; light/dark themes

Deferred past MVP

- Sync & real-time collaboration
- Live page embeds (the stable block IDs they need *are* in the MVP format)
- OneNote (`.one`) import
- Ink recognition (ink-to-text / ink-to-math)
- Backlinks & graph view
- Tables, OCR, audio notes, web clipper
- Page templates, phone & web builds
- Math solving

Roadmap at a glance

PHASE 0 Foundations — specs & validation spikes **WE ARE HERE**

- Stack decided (Flutter + Rust, Loro, SQLite) · deep specs written · remaining: editor-engine bake-off, canvas/ink/CRDT validation spikes, license ratification

PHASE 1 MVP — the single-device notebook

- The full canvas + notebooks + text/Markdown + math + ink + the open format, on Windows/macOS/Linux

PHASE 2 Own your notes everywhere — sync, embeds & polish

- Conflict-free CRDT sync · bring-your-own backend · **live page embeds** · hardened tablet/pen · full open export · backlinks, tables, templates

PHASE 3 The switch & the network

- OneNote importer · optional ink recognition · real-time collaboration · sharing & permissions · phone & web

DESIGN DIRECTION

Ink on an open page

The interface should feel like good stationery: calm, precise, quietly excellent — never loud, never in the way. The page is the hero; chrome recedes. The brand concept is **ink** — the fountain-pen line, the mathematician's margin note — set on an **open page**.

Palette

Ink — the primary. A deep indigo, distinct from OneNote's magenta-purple, evoking considered ink on paper.



Brass — the accent, used sparingly. A warm amber evoking a pen nib.



Typography

Inter for UI & body — neutral, legible, open-licensed.
Source Serif for the optional reading mode.
JetBrains Mono for code · KaTeX / STIX Two for math.

Principles in practice

Borders and subtle tints over heavy shadows. One primary action per view. Contextual toolbars, never a fixed ribbon. Motion clarifies, never decorates. WCAG 2.1 AA by default — color never carries meaning alone.

Voice. Clear, warm, unpretentious — and precise about openness. We don't call OneNote's format "completely closed"; it's "documented but impractical to use elsewhere." Accuracy is part of the brand.

THE DECISION

The stack is chosen — with evidence and revisit triggers

Openote's requirements pulled in opposite directions: **content richness** (rich text, math, collaboration) favored web rendering; **canvas control** favored native. The stakeholder's priority order — **startup speed, cross-platform consistency, and feature richness above pen latency** — resolved it: Electron fails startup (1.5–3 s cold starts, 100–200 MB bundles), Tauri fails consistency (three webview engines, fragile WebKitGTK on Linux), and with near-native ink latency demoted to a non-goal, Qt's pen advantage stopped mattering.

DECIDED

Flutter + Rust core

Flutter/Dart UI (sub-second AOT startup, one renderer everywhere, first-class Linux, team velocity) with a Rust core for CRDT & sync. ADR-0001.

DECIDED

Loro CRDT · SQLite `onote`

Loro's movable-tree CRDT behind our own Rust API (ADR-0002); a crash-safe SQLite container with an in-file JSON mirror + lossless open-folder export (ADR-0003).

OPEN

Editor bake-off

super_editor vs appflowy_editor — two 1–2-week spikes (math chips, live Markdown, canvas boxes) decide it. ADR-0004.

Build-ready specs. The technical layer is now written to code from: the **File Format Specification** (SQLite schema, CRDT encoding, the "mirror" that keeps the container open, export mapping), the **Data Model Specification** (every block's fields, identity rules, the live-embed reference model), the **Math Input Specification** (the linear build-as-you-type grammar), and the **Ink Data Specification** (stroke encoding, InkML interchange) — plus five ADRs, each with explicit revisit triggers.

The whole strategy

OneNote is beloved for four things nobody else combines — a freeform canvas, real ink, a notebook hierarchy, and beautiful math entry — and resented for four things Microsoft won't fix: a locked format, a mandatory cloud, uneven cross-platform parity (no Linux), and subscription gates on core features. **Openote's entire strategy is to match the first four with open technology and eliminate the second four by construction.** The importer that frees trapped notebooks, the open documented format, the Linux client, and local-first sync that doesn't corrupt are, together, a reason to switch that no existing product offers.

Openote — Product & Design Overview. Draft v0.2 · 22 July 2026 · Planning phase. Companion to the full documentation set: Product Vision · OneNote Teardown & Gap Analysis · Product Requirements · Technology & Framework Evaluation · Architecture Overview · Style Guide & Design System · File Format, Data Model, Math Input & Ink Data Specifications · ADRs 0001–0005. Openote is not affiliated with or endorsed by Microsoft; "OneNote" and "Microsoft" are trademarks of Microsoft Corporation, referenced for comparison and interoperability only.